



LÉONARD Cédric

PROFESSIONAL AND ASSOCIATIVE EXPERIENCES

Since October 2022, Munich // Deutsches Zentrum für Luft- und Raumfahrt (DLR)

- PhD at the Remote Sensing Technology Institute (DLR)
- "Exploration of efficient Deep Learning architectures on FPGAs for Remote Sensing"
- Application: onboard satellite data compression
- FPGA design using AMD toolchains, including Vitis AI

January – Mars 2026, Frascati // European Space Agency (ESA) - ESRIN

- Visiting Researcher at ESA Φ-lab
- "SAR RCMC data compression using hyperprior autoencoders."
- Research sharing in multiple research groups
- Networking and collaboration

June – August 2021, Rennes // Laboratory VAADER - IETR

- Internship following the 4th year research program
- "Tangled Program Graphs for Low Complexity Block Partitioning in Video Coding"
- Using a Genetic Programming technique (TPGs) to reduce the computational complexity of a video encoding standard: VVC;
- Study of TPGs hyperparameters behavior.

June - July 2020, Nantes // Naostage

- Intern at Naostage
- Handling and development of a Qt graphic library (C++): **Qaterial** ([GitHub : /OlivierLDff/Qaterial](https://github.com/OlivierLDff/Qaterial));
- Conception of PsnViewer, an application allowing real-time monitoring of **PosiStageNet** packets circulating on the network.

2017 - 2021, INSA de Rennes // Amicale de l'INSA de Rennes

- Member of the Sound&Light club (Presidency 2019 - 2020)
- Setup and technical support on every internal event at the university (festivals, concerts, shows, conferences, etc.);
- Design and implementation of lighting effects with **GrandMa2**.

2016 and 2019-2020

- Tutoring
- Tutor for a support group of 2nd year students;
- Mathematics and physics courses for high school students.



08/03/1999

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81675, Munich

Google Scholar



MISCELLANEOUS SKILLS

- Programming ([Github](#)):



- Teamwork and communication
- Scientific literature review
- Popularization and teaching

LANGUAGES

- French: Native
- English: C2
- German: B2

HOBBIES & INTERESTS

- All kinds of mountain activities
- 5 years of theater
- 13 years of judo - black belt
- Oversharing my love for science

EDUCATION

- **2021 / 2022**: Double Degree, Master of Science (Technology) Åbo Akademi University, Finland
- *Master's Thesis: GAN / REST API*
- **2017 / 2021**: Electronics and Industrial Computer Engineering, INSA de Rennes, France
- *Research program: TPG / VVC*
- **2013 / 2017**: High school diploma, focus on engineering sciences



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SCIENTIFIC TALKS & CONTRIBUTIONS

April 2026 // ACM Computing Surveys

- Journal paper
- “FPGA-Enabled Machine Learning Applications in Earth Observation: A Systematic Review”
- Systematic survey paper over RS, ML, and FPGA research
- [ACM CSUR](#)

April 2025, Vienna & Innsbruck // LPS 2025

- Conference talk (
- “Onboard Machine Learning-based Compression of Synthetic Aperture Radar (SAR) Images Using FPGA/MPSoC Hardware”
- Deploying a joint denoising and compression method on the edge
- [DLR elib](#)

April & October 2024, Vienna & Innsbruck // EGU & MARS 2024

- Poster (Natural Hazards) & Oral presentation
- “SAR GRD Data Compression using Mean-Scale Hyperprior architectures”
- Benchmarking autoencoder-based architectures on simplistic EO datasets: EuroSAT and EuroSAT-SAR
- [EGU 2024](#)

September 2023, Leeds // FRINGE 2023

- Poster presentation
- “A Benchmark for SAR Learned Data Compression On-Board”
- Exploring lightweight architectures and data format to tackle SAR SLC data compression
- [DLR elib](#)

January – June 2022, Turku // Åbo Akademi

- Master's Thesis in Computer Engineering
- “Performance exploration of the REST API using online GAN”
- Literature review about Deep Neural Networks, Generative Adversarial Networks, **REST APIs**, and Performance Testing
- Exploring the possibilities of automated test generation of a new Online learning algorithm based on **GANs**
- [Åbo Akademi doria](#)

AWARDS AND GRANTS

January – March 2026, Research Stay at ESA, Frascati, Italy

- DAAD Forschungstipendien für Doktorandinnen und Doktoranden, 2025
- ~5.000€ for the duration of the stay.